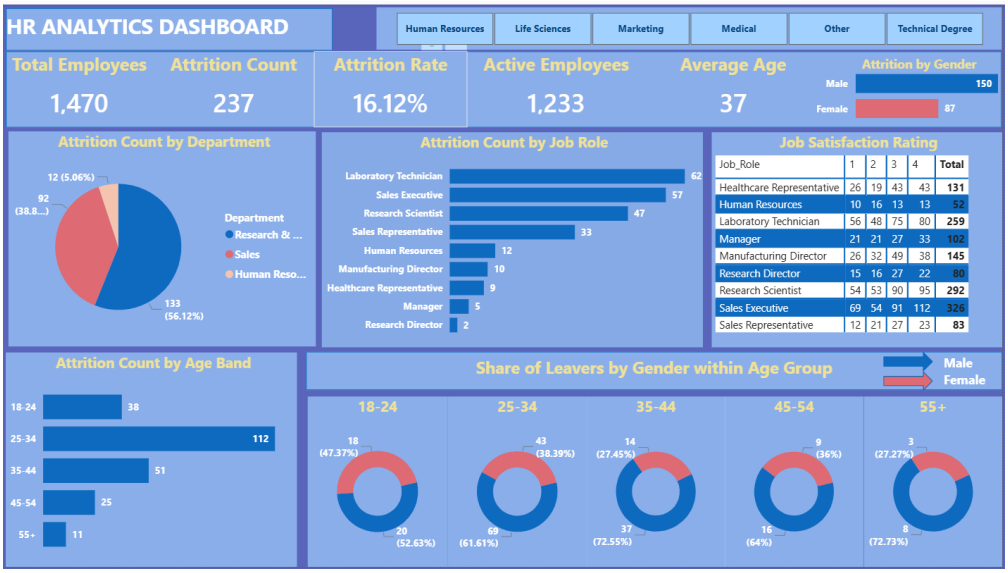
HR Employee Attrition Analysis

Key Results, Insights, and Recommended Actions



1. Executive summary

The dashboard records **237 leavers out of 1,470 employees**, giving an overall attrition rate of **16.12%**. R&D contributes the largest number of exits, while Laboratory Technicians, Sales Executives, and Research Scientists together account for **70.0%** of all leavers. These are volume indicators; group-specific attrition rates are needed to identify the highest-risk teams.

2. Workforce KPIs

Metric	Value
Total employees	1,470
Employees who left	237
Active employees	1,233
Overall attrition rate	16.12%
Average age	37 years
Male leavers	150 (63.3% of leavers)
Female leavers	87 (36.7% of leavers)

3. Key insights

Department: Research & Development recorded 133 exits (56.1% of all leavers), Sales 92 (38.8%), and Human Resources 12 (5.1%). R&D is the largest contributor to exit volume; department headcounts are needed to compare attrition rates fairly.

Job role: The largest exit counts were Laboratory Technician 62 (26.2%), Sales Executive 57 (24.1%), and Research Scientist 47 (19.8%). Together, these roles account for 166 of 237 exits (70.0%).

Age band: Employees aged 25–34 accounted for 112 exits (47.3%), followed by ages 35–44 with 51 (21.5%) and ages 18–24 with 38 (16.0%). The 25–34 band is the largest contributor by count; age-band headcounts are required to compare risk.

Gender: There were 150 male and 87 female leavers. These represent 63.3% and 36.7% of leavers, respectively. This is the composition of exits, not the attrition rate for each gender.

Job satisfaction: The dashboard's satisfaction table has 326 records for Sales Executives, 292 for Research Scientists, and 259 for Laboratory Technicians. These are row totals across satisfaction ratings, not leaver counts or attrition rates.

4. Recommended actions

1. Prioritize high-volume areas — Review exit-interview themes, workload, manager support, and development opportunities in R&D and the three roles contributing 70.0% of exits.

2. Add group attrition rates — Show both exit counts and attrition rates by department, role, age band, and gender. Rate = leavers in group ÷ total employees in group × 100.

3. Check early-tenure exits — Use the SQL tenure analysis to identify whether exits cluster in the first 1–3 years; if they do, review onboarding, role expectations, and early-career support.

4. Validate work and pay signals — Compare attrition rates with overtime, business travel, income, promotion timing, and satisfaction results. Treat these as potential signals, not proven causes.

5. Monitor outcomes — Track overall attrition and rates for priority teams monthly or quarterly, using a consistent reporting period and definition.

5. SQL analysis covered

• Attrition by department, role, gender, marital status, overtime, and business travel.

• Leavers versus stayers by income, hourly rate, salary hike, tenure, promotion timing, and manager tenure.

• Attrition rates by income band, tenure band, stock-option level, satisfaction score, training frequency, and performance rating.

• Role-level attrition-rate ranking within each department.

6. Limitations and data checks

• The dashboard shows exit counts for several groups, but not the total headcount in each group; rate comparisons cannot be concluded from counts alone.

• The dashboard screenshot does not show results for every SQL analysis, so no numerical conclusions are assigned to those queries here.

• The reporting period and source-data provenance are not specified in the supplied files.

• Back up hr_attrition_raw before running schema.sql, because the script drops the staging table after creating the four analysis tables.

7. Conclusion

The immediate priority is to investigate the teams and roles contributing the most exits, while adding group-specific attrition rates to avoid misleading comparisons. The SQL analysis can then test whether tenure, work conditions, compensation, or employee-experience measures are associated with higher attrition. Any proposed cause should be validated with additional analysis and employee feedback.